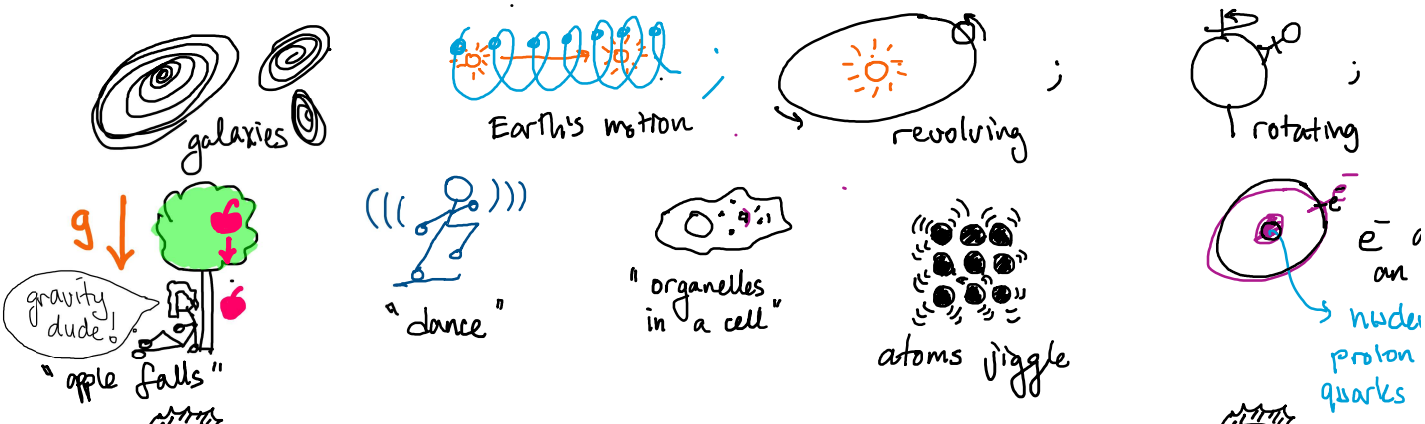
# 02-Motion in 1d,2d,3d

Tuesday, October 31, 2023 11:43 PM

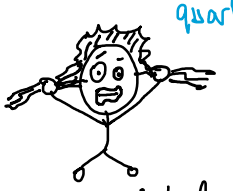
Chapter 2 + Chapter 3 = MOTION

1 dim  2dim; 3dim

WHY is motion IMPORTANT? Because EVERYTHING in the UNIVERSE MOVE

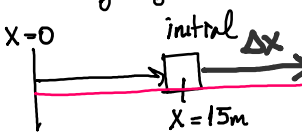


**NOTHING is EVER STATIONARY!!**



OK! Let's go back to our lives!

to define motion ⇒ start with location (you need a coord. sys.)
how location changes with time?




change in location = $\boxed{\Delta X = X_f - X_i \quad = \text{DISPLACEMENT}}$ * * YER DEĞİŞTİRME

$\boxed{\text{Delta} = \Delta = \text{final} - \text{initial}}$ * *

X is a vector but we dropped the because we are in 1 dimension.

## HOW FAST DISPLECEMENT HAPPENS? ⇒ VELOCITY

(HIZ)
$$\text{velocity} = \frac{\text{Displacement}}{\text{change in time}} = \left[\frac{m}{s}\right]$$

velocity

average velocity  instantaneous velocity

velocity ($v$) is a vector!

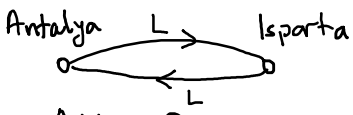
$$\left\{\text{Speed (sürat)} = \frac{\text{total distance}}{\text{total time}}\right\}$$

$$\boxed{\bar{v} = \frac{\Delta X}{\Delta t}} \left[\frac{m}{s}\right]$$

$$\boxed{v = \frac{dx}{dt}} \left[\frac{m}{s}\right]$$

deriv with

{d ≡ very little change c




$\Delta X = 0$

$v = 0 = \frac{\Delta x}{\Delta t}$

speed $= \frac{2L}{t} > 0$

speed is always positive

speed ≥ 0

* instantaneous velocity is different than a
* they are the same under certain condition

ex) an objects location vs time is given as:

| $t(s)$ | $X$ (m) |
|---|---|
| 0.00 | 0.00 |
| 1.00 | 19.0 |
| 4.00 | 277 |
| 5.00 | 200. |

3 s.f. sensitivity

a) what's average velocity between 0s and 1s?

$$\bar{v} = \frac{x_f - x_i}{t_f - t_i} = \frac{19-0}{1-0} = 19\,m/s$$

b) what's $\bar{v}$ between 1s and 4s

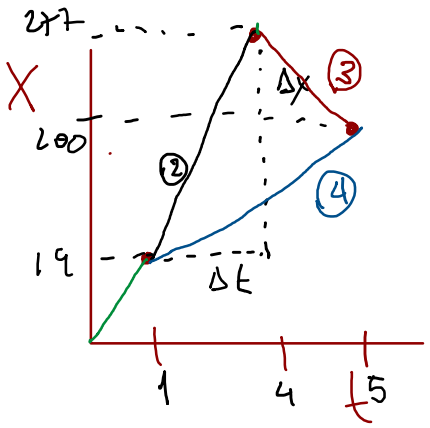
$$\bar{v} = \frac{277-19}{4-1} = 86.0\,m/s$$

c) what's $\bar{v}$ between 4s and 5s

$$\bar{v} = \frac{200-277}{5-4} = -77.0\,m/s$$

d) what's $\bar{v}$ between 1s and 5s

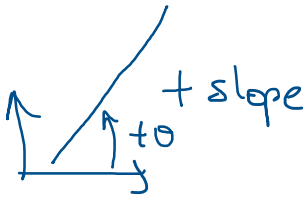
$$\bar{v} = \frac{200-19}{5-1} = 45.3\,m/s$$

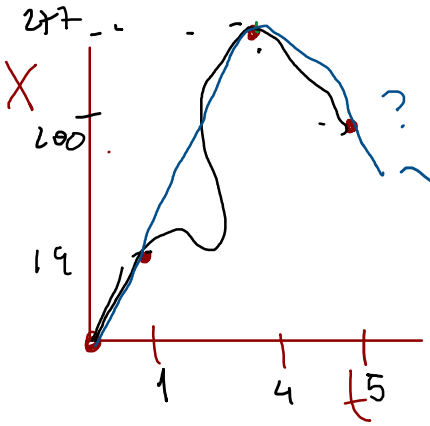



$$\bar{v}_{1s,5s} = \frac{200-19}{5-1} =$$

$$\frac{\Delta x}{\Delta t} = \text{slope}$$

$\underline{\underline{\text{eqiu}}}$






$\underline{\underline{X(t)}}$ ⇒ you can calculate $v =$

$v$ vs $\bar{v}$ ?

$$v = \underbrace{\lim_{\Delta t \to 0} \frac{\Delta x}{\Delta t}}_{\text{definition of}} = \lim_{\Delta t \to 0} \bar{v}$$

$$X = At^{n} \to \text{integer}$$

$A$ → real no

$$\underbrace{\frac{dx}{dt} = Ant^{n-1}}$$

ex) $X(t) = 20 + 5t^2$ (what's the unit of 5?)

$X = [m] = 20\,m +$

$\bar{v}$ between 1s 2s?

$$\bar{v} = \frac{\Delta x}{\Delta t} = \frac{X(2) - X(1)}{2-1} = \frac{40-25}{1} = 15\,m/s$$

$20\,m +$

$\bar{v}$ between 1s, 1.1s?

$$\bar{v} = \frac{x(1.1) - x(1)}{1.1 - 1} = \frac{26.05 - 25}{0.1} = \frac{1.05}{0.1} = 10.5$$

$\bar{v}$ between 1s 1.01s?

$$\bar{v} = \frac{x(1.01) - x(1)}{1.01 - 1} = \frac{25.1005 - 25}{0.01} = 10.05 \text{ m}$$

$\bar{v}$ = between 1s 1.001s

$\bar{v}$ = = = 10.005

| $\bar{v}$ | $\Delta t$ | $t_i$ | $t_f$ |
|---|---|---|---|
| 15 | 1 | 1 | 2 |
| 10.5 | 0.1 | 1 | 1.1 |
| 10.05 | 0.01 | 1 | 1.01 |
| 10.005 | 0.001 | 1 | 1.001 |
| ↓ | ↓ | | |
| 10 | 0 | | |

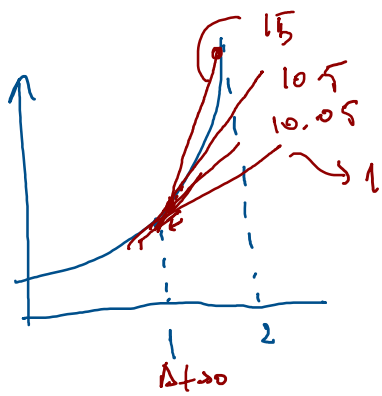
$$v = \lim_{\Delta t \to 0} \bar{v} = 10 \text{ m/s}$$

Q: what's the $v$ at t=

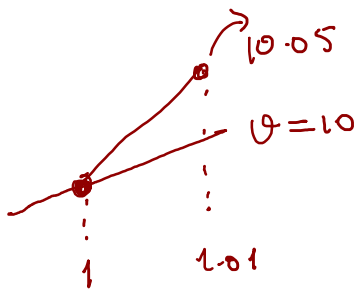
$$v = \frac{dx}{dt} = \frac{d}{dt}(20 + 5t$$

$$= 0 + 5(2)t$$

$$v\Big|_{t=1s} = 10 \text{ m/s}$$




$10 = v = \frac{dx}{dt}$ → tangent to the function at t = 1s

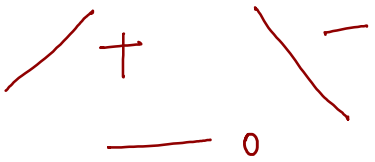


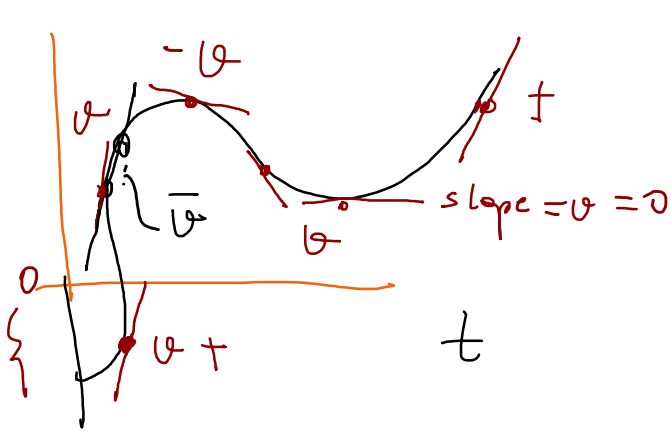

$\bar{v}$ = slopes between

$v$ = tangent at

## x-t graph

slope = velocity




$x, v \equiv$ vectors !

$x \Leftrightarrow \vec{x}$

$(+, -)$

$\underline{\underline{V}}$

Ch 2 1d motion

change in velocity with time $\equiv$ acceleration = iv

$$\bar{a} = \frac{\Delta v}{\Delta t}$$

$$a = \frac{dv}{dt}$$

$v \equiv$ instan velo

$$\bar{a} = \frac{v_f - v_i}{t_f - t_i}$$

$$= \frac{d}{dt}\frac{dx}{dt} = \frac{d^2x}{dt^2}$$

unit analysis $\underline{\underline{a}}$

$= \frac{dv}{dt}$ derivati of v with

$$a = \frac{\Delta v}{\Delta t} = \left[\frac{m/s}{s} = \frac{m}{s^2}\right]$$

$a = \frac{d^2x}{dt^2} = 2^{nd}$ degree location

$$\boxed{a = \lim_{\Delta t \to 0} \frac{\Delta V}{\Delta t} = \lim_{\Delta t \to 0} \bar{a} = \frac{dv}{dt}}$$

instantaneous acc.

$x = 20 + 5t^2$

$v = 10t$

$a = \frac{d}{dt} v = 10\ m/s$

ex) $v = 60\frac{m}{s} + \underbrace{0.5\frac{m}{s^3}t^2}_{\frac{m}{s}}$

a) $\bar{a} = ?$ $t_i = 1s$ $t_f = 3s$ $\Rightarrow$ $\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v(3) - v(1)}{3-1}$

$= \frac{64.5 - 60.}{2}$

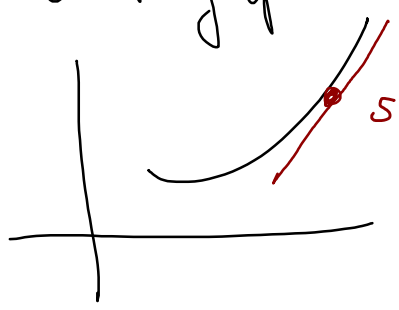
| | $t_i$ | $t_f$ | $\bar{a}$ |
|---|---|---|---|
| $\bar{a}$ | 1 | 1.1 s | 1.05 $m/s^2$ |
| | 1 | 1.01s | 1.005 $m/s^2$ |
| | ⋮ | ⋮ | ↓ |
| | ⋮ | $\Delta t \to 0$ | $1\ m/s^2$ |

$\lim_{\Delta t \to 0} \bar{a} = a =$

$a = \frac{dv}{dt} = \frac{d}{dt}(60 + 0.5t^2) = 0.5(2)t^1 = 1t$

$a(t = 1s) = 1\ m/s^2$

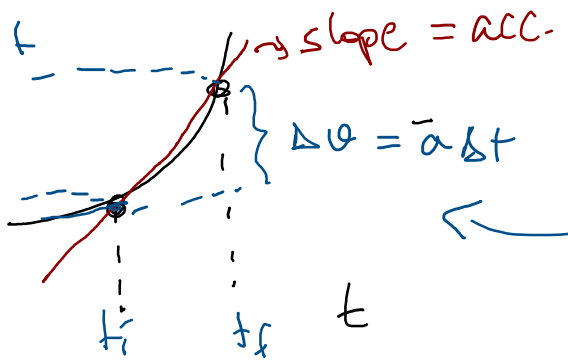
v-t graph




slope = acceleration

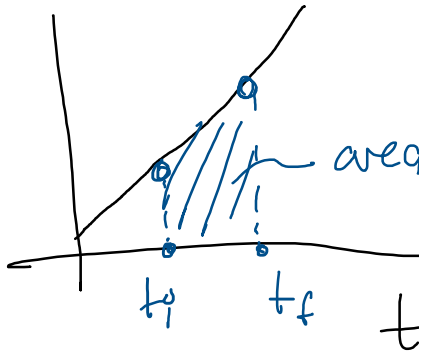
$a = \frac{\Delta v}{\Delta t}$

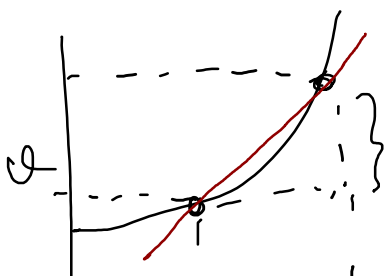
$a\Delta t = \Delta v$




slope

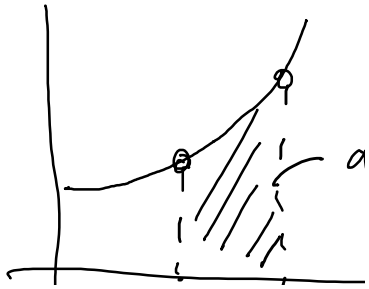
area

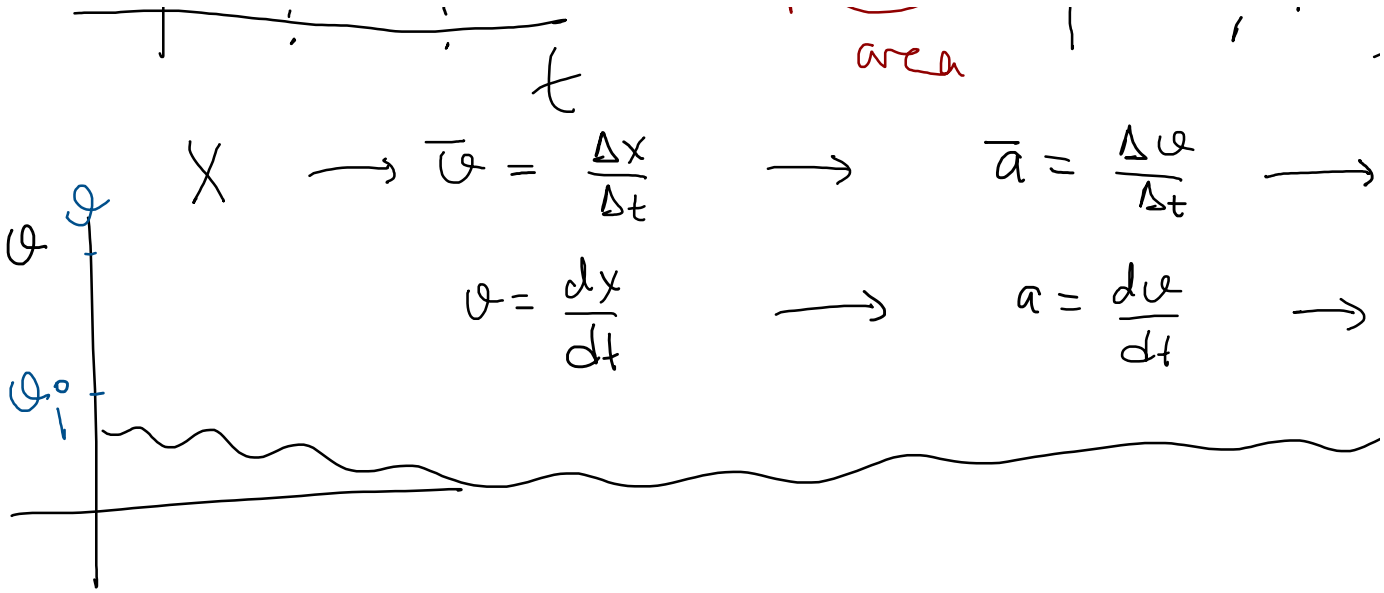



$\boxed{\text{slope} \Rightarrow \bar{v}}$

slope

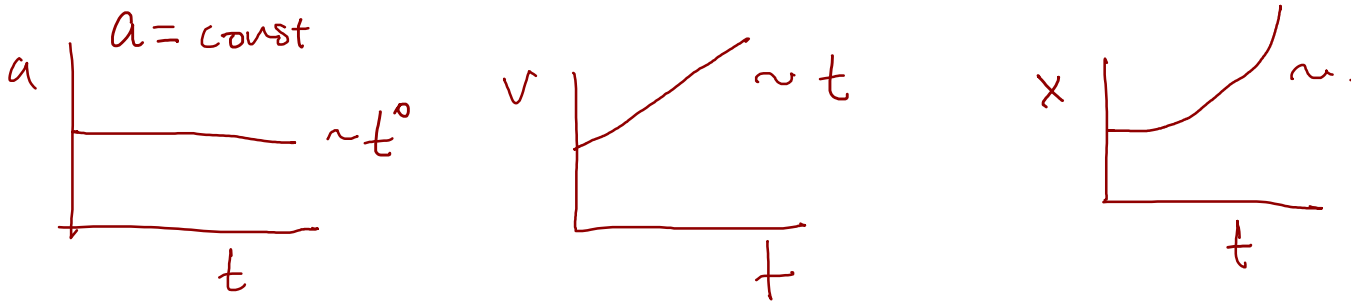


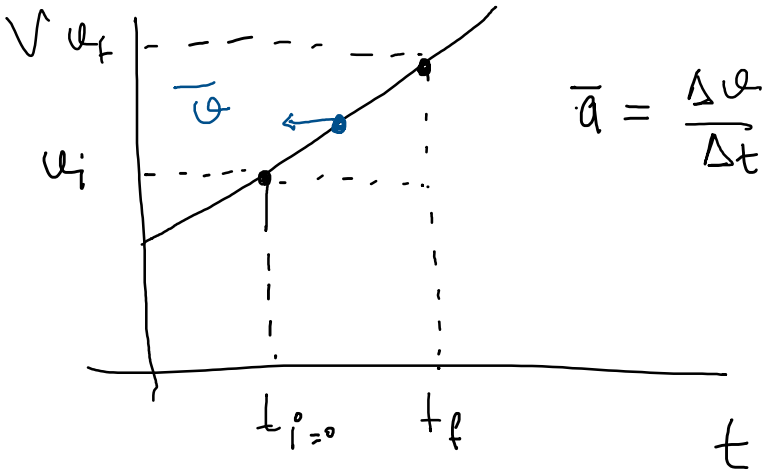



$$X \longrightarrow \bar{v} = \frac{\Delta x}{\Delta t} \longrightarrow \bar{a} = \frac{\Delta v}{\Delta t} \longrightarrow$$

$$v = \frac{dx}{dt} \longrightarrow a = \frac{dv}{dt} \longrightarrow$$

constant acceleration situation:




$$\bar{a} = \frac{\Delta v}{\Delta t} = \frac{v_f - v_i^0}{t_f - t_i} = a = \frac{v_f - v_i^0}{t - 0}$$

$\boxed{t_i = 0}$

$\boxed{t_f = t}$

$v_f - v_i = a$

(1) $v_f = v_i^0 +$

$v(t) = v_i$

$$\bar{v} = (v_i^0 + v_f)/2 = \frac{\Delta x}{\Delta t} = \frac{x_f - x_i^0}{t - 0}$$

$$x_f - x_i = (v_i^0 t + v_f t)\frac{1}{2}$$

$$x_f - x_i^0 = \left(v_i^0 t + (v_i + at)t\right)\frac{1}{2} = \frac{1}{2}\left(2v_i t + a\right.$$

(2) $$\boxed{x_f = x_i^0 + v_i^0 t + \frac{1}{2}at^2}$$ **

$\boxed{v_f = v_i -}$

$$\boxed{X(t) = X_i + v_i t + \frac{1}{2} a t^2}$$

③ free of time $\Rightarrow$ ① $t = \dfrac{v_f - v_i}{a} \Rightarrow$ ②

$$X_f = X_i + v_i \left(\frac{v_f - v_i}{a}\right) + \frac{1}{2}\cancel{a} \frac{(v_f - v_i)^2}{a^{\cancel{2}}}$$

$$= X_i + \frac{v_i v_f}{a} - \frac{v_i^2}{a} + \frac{v_f^2}{2a} + \frac{v_i^2}{2a} - \frac{\cancel{2} v_f}{\cancel{2}}$$

$$-\frac{v_i^2}{2a}$$

$$X_f = X_i - \frac{v_i^2}{2a} + \frac{v_f^2}{2a} \rightarrow \boxed{v_f^2 = v_i^2 + 2a(X_f}$$

constant acc. $\underline{\underline{a}}$

$\{ v_f = v_i + at$ ② $\{ X_f = X_i + v_i t + \frac{1}{2} a t^2$ ③ $\{ v_f^2 = v_i^2 + 2$

$i$ = initial
$f$ = final

$v_f, v_i, X_f, X_i, a$ $\rightarrow +$, $\leftarrow -$ $\quad \underline{\underline{1d}}$ $\quad +$

ex) $\rightarrow a = 4\,m/s^2$ @ $t = 0s$ $\quad X_i = 5m$ $\quad v_i = 15\,m/s$

a) at $t = 2s$ $\quad X_f = ?$ $\quad v_f = ?$

$X_f = 5 + 15(2) + \frac{1}{2} 4\, 2^2 =$

$v_f = 15 + 4(2) = 23\,m/s$

b) when $v = 25\,m/s$ what's its location? $\underline{X_f = ?}$

$25^2 = 15^2 + 2(4)(X_f - 5) \Rightarrow X_f = 55\,m$

2nd way find time $\Rightarrow t = ?$ @ $v_f = 25\,m/s$

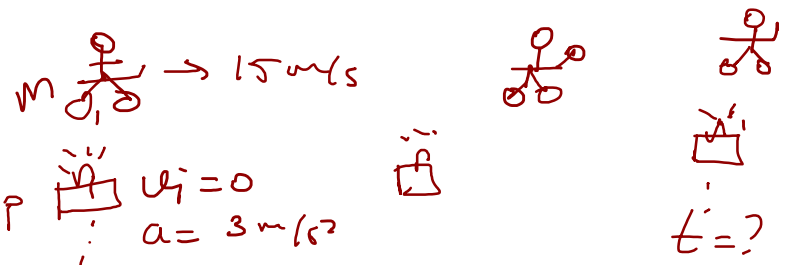
① $25 = 15 + 4t \Rightarrow t = 10/4 = 2.5s \longrightarrow X_f = 5.$

ex) a motorist traveling at const 15 m/s
police car which is stationary. After passing pc
tries to catch the motorist with const. acc. a

a) when does the police catch the motorist?

b) what is the police cars velocity when police catc

c) How far does each vehicle travelled?




$x_{mf} = \cancel{x_{mi}}^{0} + v_{mi}$

equal

$x_{pf} = 0 + 0t$

$x_{mi} = 0 \quad a_m = 0$

$x_{pi} = 0$

$v_{pi} = 0$

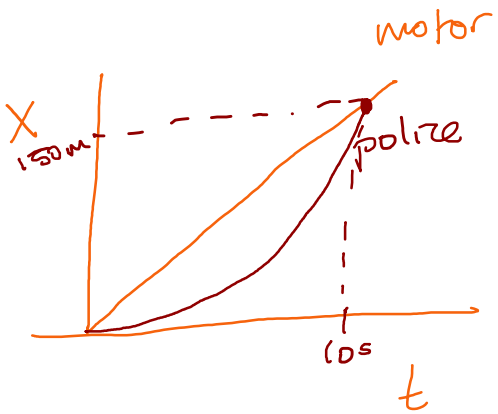
$x_f = x_{pf} = x_{mf} = 15t$

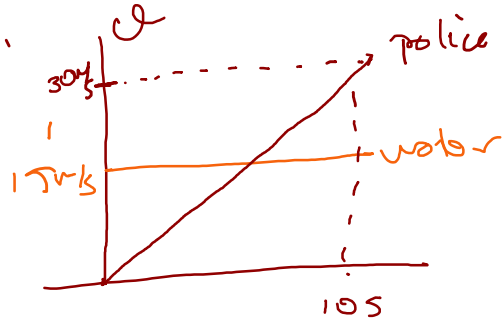
$t =$

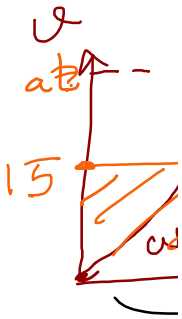
b) $v_{pf} = \cancel{v_{pi}}^{0} + a_p t = 3(10) = 30\ m/s > 15\ m/s$

c) $x_f = x_{pf} = x_{mf} = 15t = \frac{3}{2}t^2$ (10, 10)

$= 150\ m = 150\ m$



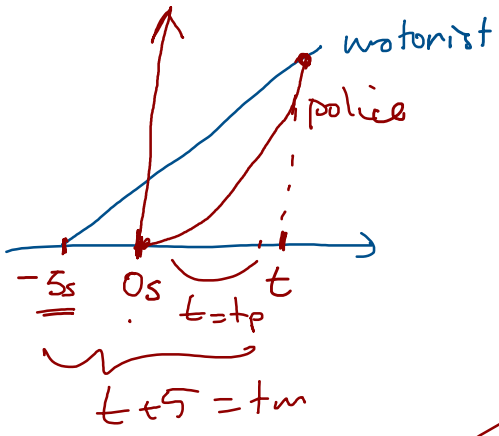

$x_p = \frac{3t^2}{2}$

same question but police starts moving after

motorist passes the police.

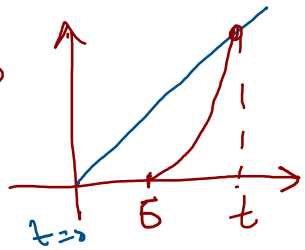
$$x_f = x_i + v_i t + \frac{1}{2} a t^2$$

$$\begin{cases} t+5 = t_m \\ t = t_p \end{cases}$$




$$x_m = 15(t + 2$$

$$\begin{pmatrix} t_m = t \\ t_p = t-5 \end{pmatrix}$$




not save

$$\Rightarrow x_m = 15t = x_p = \frac{1}{2} a (t-5$$

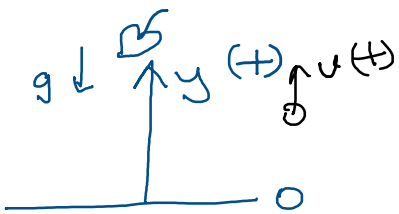
$$\{ At^2 + Bt + C = 0 \rightarrow t = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$$

$$t = \frac{15 \pm \sqrt{15^2 + 4(1.5)\,75}}{2\,(1.5)} \Rightarrow \frac{5 \pm \sqrt{85}}{} \quad \frac{15 \pm 25.98}{3}$$

tie for

13.66 t

FREELY FALLING OBJECTS



$X \rightarrow Y$

$a \rightarrow (-g)$

$v \rightarrow v_y \qquad a \rightarrow -g \qquad X \rightarrow Y$

+y

$$v_{yf} = v_{yi} - gt \qquad \left\{ \begin{array}{l} x_f = x_i + v_i t + \frac{1}{2} a t^2 \\ y_f = y_i + v_{yi} t - \frac{1}{2} g t^2 \end{array} \right\}$$

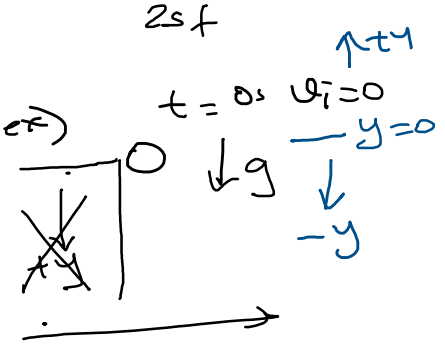
$g = 9.8\ m/s^2 = 9.81\ m/s^2 = 9.80\ m/s^2$

2sf — 3sf. in book

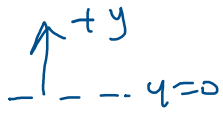
$y_f = \cancel{y_i}^{0} + \cancel{v_i}$

$v_f = \cancel{v_i}^{0} - g$

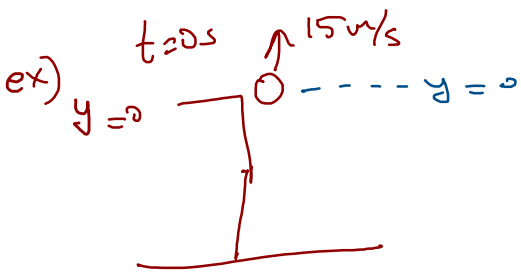
ex)




| t(s) | $y_f$ (m) | $v_f$ (m/s) |
|---|---|---|
| 1 | -4.9 | -9.8 |
| 2 | -19.6 | -19.6 |
| 3 | -45.1 | -29.4 |




ex)




a) @ t=1s $y_f = ?$ $v_f = ?$ $v_{yf} =$

$y_f = \cancel{y_i}^{0} + v_i t - \frac{1}{2} g t^2 = 15(1$

$v_f = v_i - gt = 15 - 9.8(1$

b) @ t=4s $y_f = ?$ $v_f = ?$

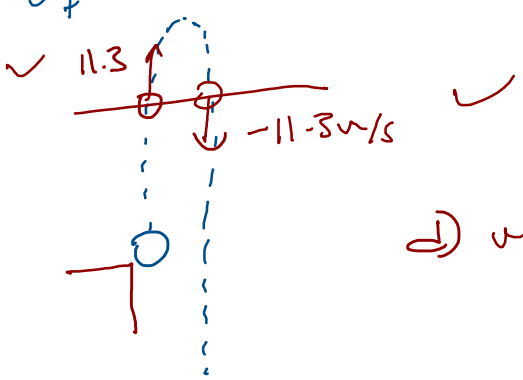
$y_f = 0 + 15(4) - 4.9(4)^2 =$

$v_f = 15 - 9.8(4) = -24.2$

c) what's the ball's velocity when its height is

$v_f^2 = v_i^2 - 2g(y_f - y_i)$

$v_f^2 = 15^2 - 2(9.8)(5 - 0) \Rightarrow v_f^2 = 127$




d) what's the ball's max

$v_f^2 = v_i^2 - 2g(y_f - y_i)$ ($y_f \to$ max, $y_i \to 0$)

$0 = 15^2 - 2(9.8)\, y_{max}$

$v_i = 15.0\ m/s$

$g = 9.80\ m/s^2$

e) what's the ball's acceleration at max heig

$a = -g = -9.8\ m/s^2$

f) why does it

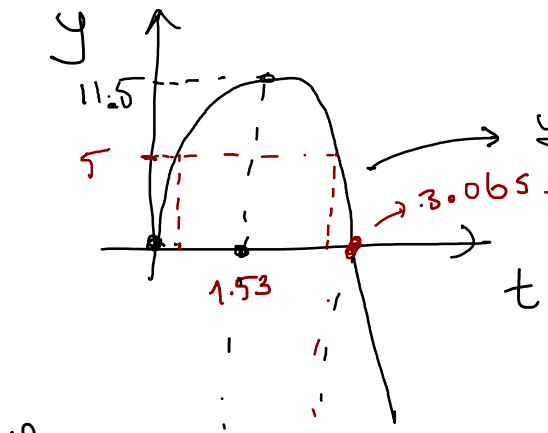
e) when does ...

$v_f = v_i - gt$

$\downarrow$

$0 = 15 - 9.8t$

$t = 1.53\,s$



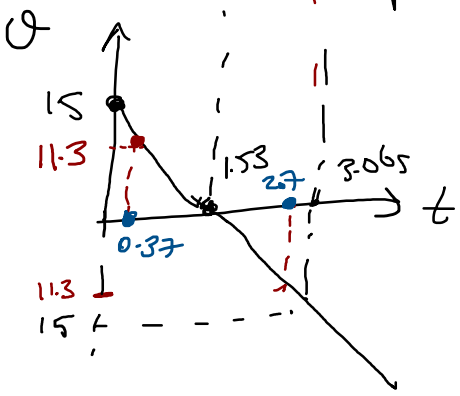

$y = 15t - 4.9t^2$

$v_f = v_i - gt$

$\downarrow$

$-15 = 15 - 9.8t$

$t = \frac{30}{9.8} = \underline{\underline{3.06\,s}}$

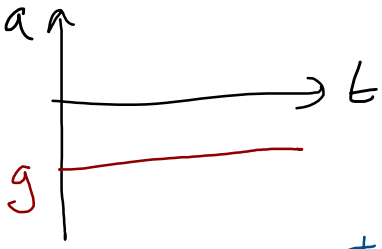



f) At what <u>times</u> the ball

$v_f = v_i - gt$ $\quad \{ y_f = y_i$

$5 = 0 -$

$4.9t^2 - 15t +$




$\frac{15 \pm \sqrt{127}}{9.8} \to 0.38,\ 2.68$

$= \frac{-(-15) \pm \sqrt{15^2 -}}{2(4.9)}$

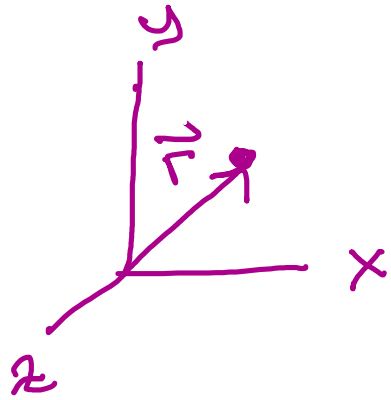
1d motion $\underline{a}$ = const. $v_f = v_i + at$

$x_f = x_i + v_i$

$v_f^2 = v_i^2 + 2$

---

Motion in 2d & 3d CH 3




* $\vec{r} = x\hat{i} + y\hat{j} + z\hat{k}$ (3d)

$\downarrow$ location

$\frac{d\vec{r}}{dt} = \vec{v} =$ ins

Inst velocity $\Rightarrow$ $\vec{v} = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} + \frac{dz}{dt}\hat{k}$

$\vec{v}$

$\vec{\bar{v}} = \frac{\Delta x}{\Delta t}\hat{i} + \frac{\Delta y}{\Delta t}\hat{j} + \frac{\Delta z}{\Delta t}\hat{k} = \bar{v}_x\hat{i} + \bar{v}_y\hat{j} + \bar{v}_z\hat{k}$ (

acceleration $\Rightarrow$ ave $\rightarrow$ $\vec{a} = \frac{\Delta v_x}{\Delta t}\hat{i} + \frac{\Delta v_y}{\Delta t}\hat{j} + \frac{\Delta v_z}{\Delta t}\hat{k} =$

instantaneous acc $\vec{a} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} + \frac{dv_z}{dt}\hat{k} =$

---

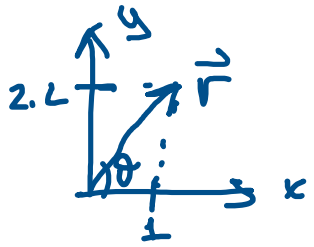
x) 2d motion $X(t) = 2 - 0.25t^2$ $Y(t) = t$

a) what is the location of the object at $t = 2s$?

$\vec{r}(2) = [2 - 0.25(2)^2]\hat{i} + [$ $(2)^3]\hat{j}$

$r(2) = [2 - 0.5(2)^2]\hat{i} + [2 + 0.025(2)^2]\hat{j}$

$|\vec{r}| = \sqrt{1^2 + 2.2^2} = 2.4\,m$ $\qquad \theta = \tan^{-1}\left(\frac{2.2}{1}\right) =$



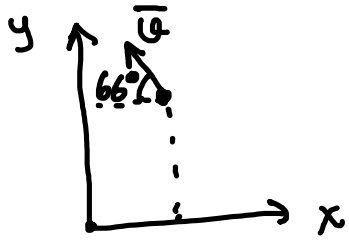

b) what's the average vel between 0s and 2s?

$$\vec{v} = \frac{\Delta x}{\Delta t}\hat{i} + \frac{\Delta y}{\Delta t}\hat{j} = \frac{x(2) - x(0)}{2 - 0}\hat{i} + \frac{y(2) - y(}{2 - 0}$$

$$= \frac{1 - 2}{2}\hat{i} + \frac{2.2 - 0}{2}\hat{j}$$

$$|\vec{v}| = \sqrt{0.5^2 + 1.1^2} = 1.2\ m/s$$

$$\theta = \tan^{-1}\left(\frac{1.1}{-0.5}\right) \approx -66^\circ$$




b) $\vec{v}(2) = ?$ inst. vel. at time = 2s? $\qquad x(t) =$

$$v = \frac{dx}{dt}\hat{i} + \frac{dy}{dt}\hat{j} = \underbrace{-2(0.25)t}_{v_x}\hat{i} + \underbrace{(1 + 0.0}_{v_y}$$

e

$$\vec{v}(2) = -\hat{i} + 1.3\hat{j}\ m/s \qquad |\vec{v}$$

c) $\vec{a}$ (ave. acc) between 0s and 2s?

$$\vec{a} = \frac{\Delta v_x}{\Delta t}\hat{i} + \frac{\Delta v_y}{\Delta t}\hat{j}$$ inst. vel. $\qquad v(t) = -0.5$

$$= \frac{v_x(2) - v_x(0)}{2 - 0}\hat{i} + \frac{v_y(2) - v_y(0)}{2 - 0}\hat{j} = \frac{-1 - 0}{2}\hat{i}$$

$= -0.5\,\hat{j}$

$|\vec{a}| = \sqrt{0.5^2 + 0.15^2}\ m/s^2$ ; $\theta = \tan$

$0.52\ m/s^2$ $\theta = -1$

d) $\vec{a}(t=2s) = ?$ inst. acc. $t=2$.

$\vec{a} = \frac{dv_x}{dt}\hat{i} + \frac{dv_y}{dt}\hat{j} = -0.5\,\hat{i} + 0.15t\,\hat{j} = \vec{a}(t)$

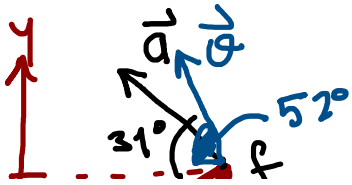
$|\vec{a}(2$

e) Show on graph the location of the
the inst. vel. "
the inst. acc. "

$\vec{r}(2) = \hat{i} + 2.2\hat{j}$

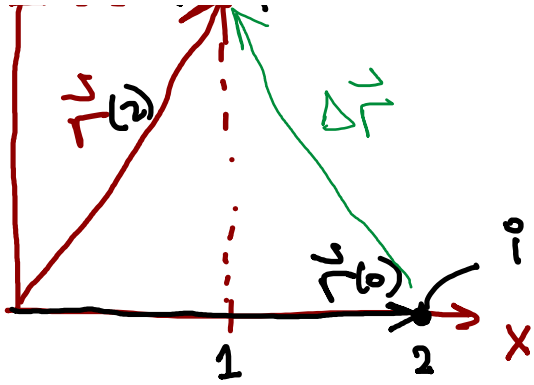
$\vec{v}(2)$

$\vec{a}(2)$




$\vec{v}(2) = (1.$

$\vec{a}(2) = (0.5$



$\Delta\vec{r}$ is somewhat in $\vec{v}$, i

motion 2d & 3d

* If an object follows a curve path its

$\Delta x, \Delta r = 0$ sae location $\bar{a} = 0$

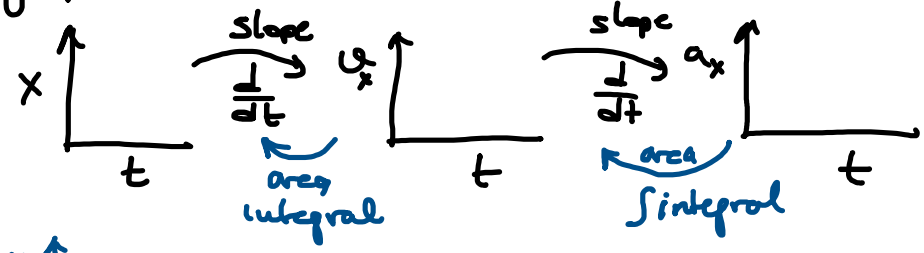
special case full turn.

$\vec{r}(f) - \vec{r}(i) = \Delta\vec{r} \neq 0 \Rightarrow \vec{v} \neq 0$ $\quad \vec{v}_f - \vec{v}_i \neq 0$ $\quad v_f$ $\quad v_i$ $\quad \Delta v \neq$

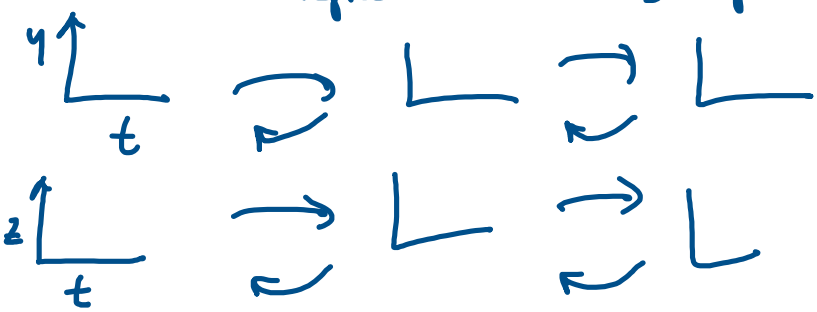
uniform circular motion ; (düzgün dairesel hareket)

$R, |\vec{v}| = \text{const}$ $\quad \Delta\vec{v} \neq 0 \Rightarrow a \neq 0$

graphs in 2d & 3d




X 3

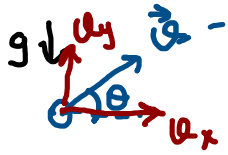


2d motion → projectile motion (eğik
→ circular motion

jectile motion (no air resistance; no spin (falso yok!))

$\downarrow g$

$\rightarrow x \quad a_x = 0$




$\vec{v}, \theta \Rightarrow v_x, v_y$

2d $\quad$ 1d + 1d

$\Rightarrow$ Pro $x_f = x_i + v_{xi} t + \frac{1}{2} a_x t^2 \quad a_x = 0; \quad x_i = 0 \quad x_f = v_{xi} t \Rightarrow$ [

y direction freely falling object

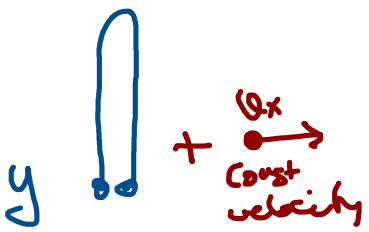
$y_f = y_i + v_{yi} t - \frac{1}{2} g t^2 \quad ; \quad y_i = 0 \Rightarrow y(t) = v_{yi} t - \frac{1}{2} g t^2$

$v_{yf} = v_{yi} - gt \qquad v_y(t) = v_{yi} - gt$

$v_{yf}^2 = v_{yi}^2 - 2g\Delta y \Rightarrow v_y^2(t) = v_{yi}^2 - 2g\,y(t)$

$(y_f - y_i)$

$v_{xi} = v_i \cos\theta \qquad v_{yi} = v_i \sin$



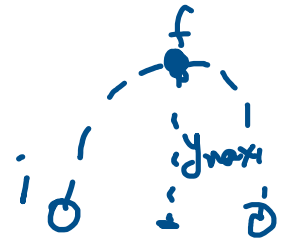
Projectile Motion

How high can it go? } y

How long will it stay in the air?

How far will it travel? } x (Range = Menzil)

How far will it travel?




$v_{yi} = v\sin\theta$

I) how long will it take to re

$t$; $y_{max}$.

$v_{yf} = v_{yi} - gt$ $\quad t = \frac{v_{yi}}{g}$

$0$

$y_f = y_i + v_{yi}t - \frac{1}{2}gt^2$

$y_i \to 0$

$y_{max} = (v\sin\theta)\frac{v\sin\theta}{g} - \frac{1}{2}$

$$\boxed{y_{max} = \frac{v^2\sin^2\theta}{2g}}$$

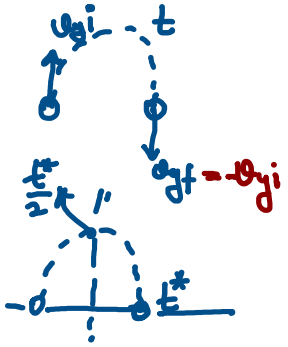
II) way

$v_{yf}^2 = v_{yi}^2 - 2g(y_{max} - y_i)$

$v_{yf} \to 0$, $y_i \to 0$

$$\frac{v_{yi}^2}{2g} = y_{max} = \frac{v^2\sin^2\theta}{2g}$$

b) How long will it stay in air = time of flight?

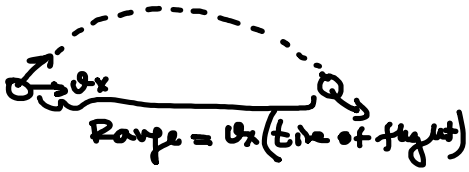



$v_{yf} = v_{yi} - gt$

$-v_{yi} = v_{yf} - gt$ $\quad t = \frac{2v_{yi}}{g} = \frac{2v\sin\theta}{g}$ = time of flight

$\frac{t}{2} \Rightarrow$ time to for max height $= \frac{v\sin\theta}{g}$

c) How far = range

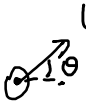


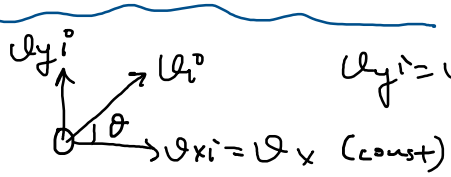

$X_{max} = v_x t$

$$\text{Range} = (v\cos\theta)\left(\frac{2v\sin\theta}{g}\right) = \frac{2v^2\sin\theta\cos\theta}{g} = \frac{v^2\sin 2\theta}{g}$$

not the range

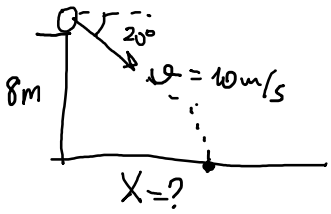
$g \downarrow$ $\quad \vec{v}_i = \vec{v}_0 = \vec{v}$




$v_{yi} = v_i \sin\theta$

$v_x = v_i \cos\theta$




horizontally

How far will the ball hit the ground?

① $y_f = y_i + v_{yi} t - \frac{1}{2} g t^2$

② $v_{yf} = v_{yi} - g t$ ③ $v_{yf}^2 = v_{yi}^2 - 2g\Delta Y$

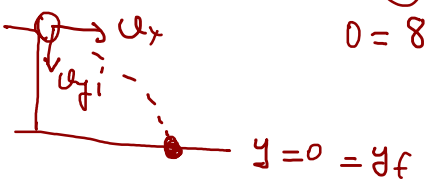
④ $X_f = v_x t$

$X = v_x t$ ④

? ✓ ?

$t$? → ② or ①

✗




$v_{yi} = 10 \sin 20 = 3.42 \, m/s$

$v_x = 10 \cos 20 = 9.40 \, m/s$

① $0 = 8 + (-3.42)t - 4.9t^2$ (+y)

$At^2 + Bt + C = 0$

$t_{1/2} = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A}$

$t = \frac{3.42 \pm \sqrt{3.42^2 + 4(4.9)8}}{-9.8}$

$t = \cancel{-1.7 s}$ $t = 0.98 s$

$X = v_x t = (9.40)(0.98) = 9.2 m$

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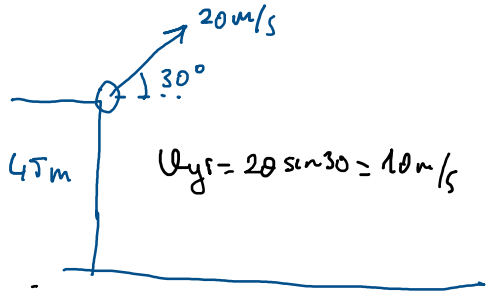



a) what is the time of flight?

1st eqn.

$0 = 45 + 10t - 4.9t^2$

$\frac{-10 \pm \sqrt{100 + 4(4.9)45}}{-9.8}$

$t \not\to t < 0$ ✗

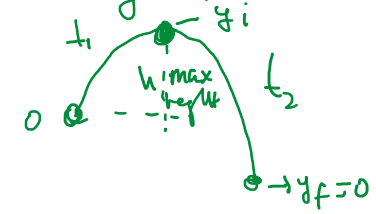
$\frac{-10 \pm \sqrt{982}}{-9.8} = 4.218 s$ ✓

① $y_f = y_i + v_{yi} t - \frac{1}{2} g t^2$

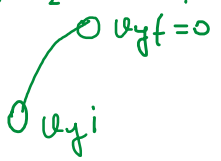
② $v_{yf} = v_{yi} - g t$ ③ $v_{yf}^2 = v_{yi}^2 - 2g\Delta Y$

④ $X_f = v_x t$

2nd way to find time of flight




$t_1 + t_2$ = time of flight




2nd formula

$0 = 10 - 9.8 t_1$

$t_1 = \frac{10}{9.8} s = 1.02 s$

3rd formula

$0 = 10^2 - 2(9.8)(y_{max} - 0)$

$h = \frac{10^2}{19.6} = 5.1 m$

1st formula

$v_{yi} = 0$

$0 = y_i + 0 - 4.9t_2^2$

$0 = 50.1 - 4.9t_2^2$

14.6

$(45+h)$ ?

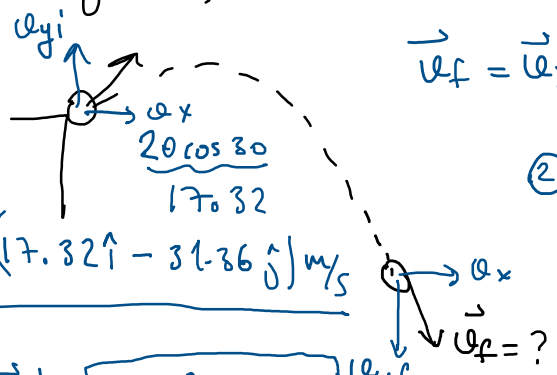
$50.1$

$t_2 = \sqrt{\frac{50.1}{4.9}} = 3.197$

time of flight

$t_1 + t_2 = 1.02 + 3.20 = 4.22\,s$ ✓

b) what's (the crash velocity) the velocity just before it hits the ground?




$\vec{v}_f = \vec{v}_x + \vec{v}_{yf}$

② $v_{yf} = v_{yi} - g\,t$ → time of flight

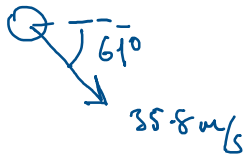
$= 10 - 9.8(4.22)$

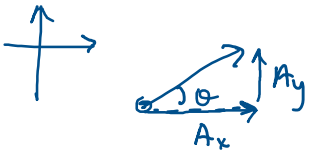
$= -31.36\ m/s$

$\vec{v}_f = (17.32\hat{i} - 31.36\hat{j})\ m/s$

$|\vec{v}_f| = \sqrt{17.32^2 + 31.36^2} \rightarrow 35.83 \simeq 35.8\ m/s$

$\theta = \tan^{-1}\left(\frac{-31.36}{17.32}\right) = -61.1^\circ \simeq -61^\circ$

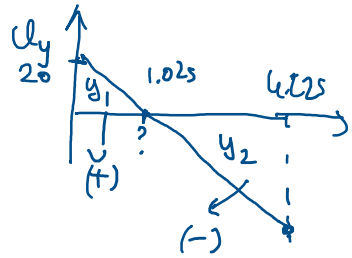


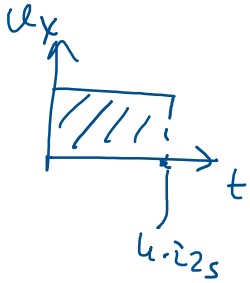




$\theta = \tan^{-1}\left(\frac{A_y}{A_x}\right)$

$A_y = -31.36$

Projectile motion

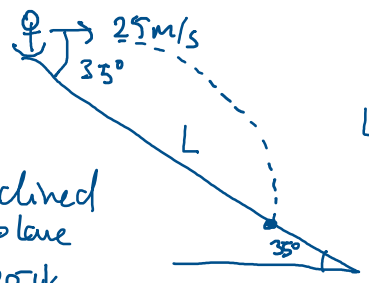





ⓒ How far it travels

$X = v_x t$

$= 17.32(4.22) = 73.1\,m$




inclined plane epic depth

L = ?

skier has 25 m/s velocity horizontal ; $v_{yi} = 0$

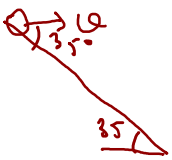
inclined has 35° inclination.

① $y_f = y_i + v_{yi} t - \frac{1}{2} g t^2$

② $v_{yf} = v_{yi} - g t$

③ $v_{yf}^2 = v_{yi}^2 - 2g\,\Delta Y$

④ $X_f = v_x t$

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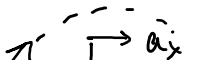
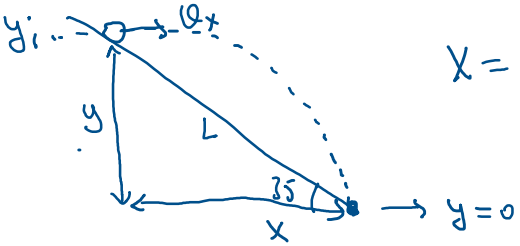


— ~~way~~




$X = L\cos 35 \longleftrightarrow y = L\sin 35$

$y_f = y_i + \cancel{v_{yi}t}^{\,0} - \frac{1}{2}gt^2$

$0 = y - \frac{1}{2}gt^2$

$\boxed{\frac{y}{?} = \frac{gt^2}{2}?}$

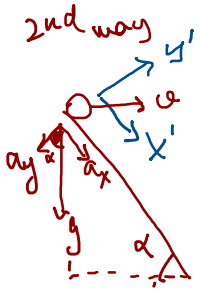
$\boxed{X = v_x t?}$

$$\frac{y}{x} = \frac{L\sin\theta}{L\cos\theta} = \tan\theta = \frac{gt^2/2}{v_x t} \Rightarrow \tan 35 = \frac{9.8t}{2(25)}$$

$$X = v_x t = 25(3.57) = 89.25 = L\cos 35 \qquad t = \frac{50\tan 35}{9.8}$$

$$\frac{89.25}{\cos 35} = 109\,m \qquad \boxed{t = 3.57\,s}$$

2nd way



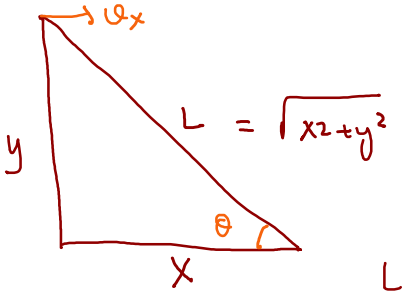

$a_y = g\cos\alpha$

$a_x = g\sin\alpha$

$X_f' = X_i' + v_{xi}'t + \frac{1}{2}a_x t^2$

$y_f' = y_i' + v_{yi}'t + \frac{1}{2}a_y t^2$

$v_{xi}' = v_i\cos\alpha \qquad v_{yi}' = v_i\sin\alpha$




$X = v_x t \qquad y = \frac{1}{2}gt^2$

$$\frac{y}{x} = \tan\theta = \frac{gt}{2v_x}$$

$$L = \left[v_x^2 t^2 + \frac{g^2t^4}{4}\right]^{1/2} \qquad t = \frac{2v_x\tan\theta}{g}$$

$$= \frac{2v_x\tan\theta}{g}\left[v_x^2 + \frac{\cancel{g^2}}{\cancel{4}}\,\frac{\cancel{4}v_x^2\tan^2\theta}{\cancel{g^2}}\right]^{1/2}$$

$L(v_x, \theta)$

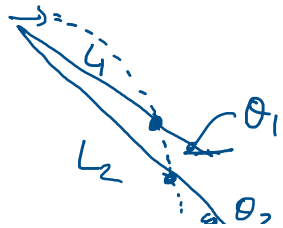
$$= L = \frac{2v_x^2\tan\theta}{g}\sqrt{1+\tan^2\theta}$$

$L\uparrow;\ v_x\uparrow;\ \theta\uparrow;$

$$= \frac{2(25)\tan 35}{9.8}\sqrt{1+\tan^2 35} = 109\,m$$

Unit analysis

$$L = [m] = \frac{m^2/s^2}{m/s^2} = [m]$$




$\theta\uparrow$

$\uparrow$

trajectory = $y(x)$ = ?

① $y(t) = y_f = y_i + v_{yi} t - \frac{1}{2} g t^2$

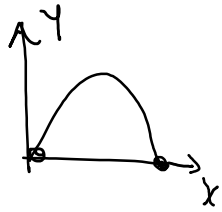
$y(t) = -at^2 + bt + c$

$y\left(\frac{x}{v_x}\right)$ $\quad y(t) = y(5t)$

② $x = v_x t \qquad t = \frac{x}{v_x}$

$$y_f = y(x) = y_i + v_{yi}\frac{x}{v_x} - \frac{1}{2} g \frac{x^2}{v_x^2}$$

$$y(x) = y_i + \cancel{v_i} \sin\theta \frac{x}{\cancel{v_i}\cos\theta} - \frac{1}{2}\frac{g}{v_i^2\cos^2\theta} x^2$$




$$y(x) = -\frac{g}{2v_i^2\cos^2\theta} x^2 + \tan\theta\, x + y_i$$

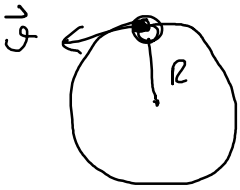
$$y(x) = Ax^2 + Bx + C$$

---

2D motion ⇒ projectile motion ✓

↳ circular motion

## Uniform Circular Motion (UCM)

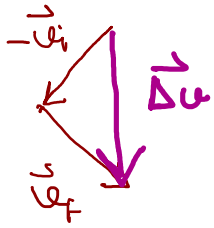



$|\vec{v}| = \text{const}$, $R = \text{const}$ } UCM

$|\vec{v}| \neq \text{const}$ Non UCM (NUCM)

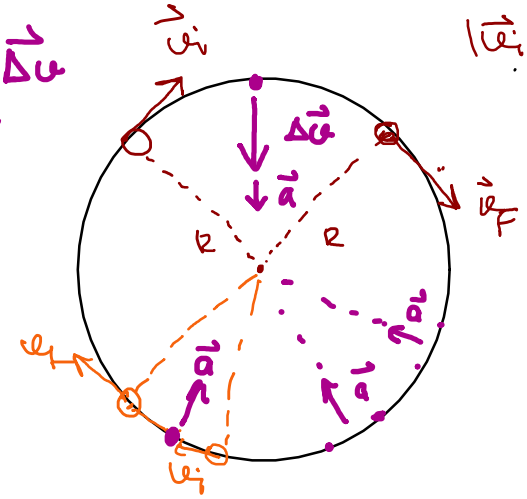
acceleration $\neq 0$ on a circular path.

$$\vec{a} = \frac{\Delta\vec{v}}{\Delta t} = \frac{\vec{v}_f - \vec{v}_i}{t_f - t_i}$$




UCM

$|\vec{v}_i| = |\vec{v}_f| =$

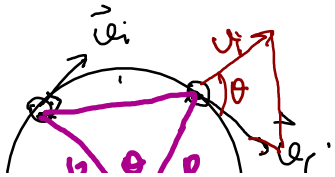



* $|\Delta\vec{v}| \neq 0 \quad |\vec{a}| \neq 0$

* $\vec{a}$ direction is always towards center.

↳ $\vec{a}$ is in radial direction

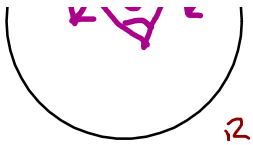
$\vec{a} = \vec{a}_r =$ rad. acc.

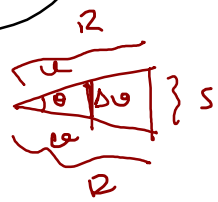


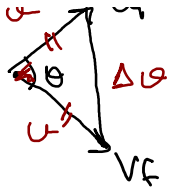

$$\vec{a} = \frac{\Delta\vec{v}}{\Delta t}$$

compare 2 triangle.

$$\frac{\Delta v}{v} = \frac{s}{R} \Big\} ?$$






similar trio

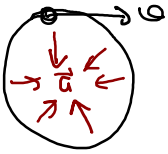
$\theta$ = same

$$\frac{\frac{\Delta v}{\Delta t}}{v} = \frac{\frac{s}{\Delta t}}{R} \qquad \frac{\text{disp.}}{\text{time}} =$$

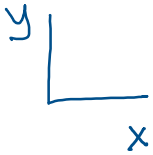
$$\frac{a}{v} = \frac{v}{R} \Rightarrow \boxed{a = \frac{v^2}{R}}$$

UCM ; radial acc.

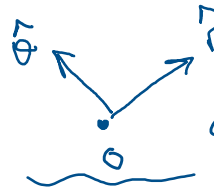
UCM



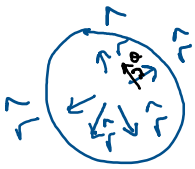

$$\Big\} \boxed{\vec{a}_r = \frac{v^2}{R}(-\hat{r})}$$




$\Longleftrightarrow$




(centrifugal acc. ≡ vertezal line)




UCM ; $v = \text{const}$ $\qquad vT = 2\pi R$

period $\qquad \frac{2\pi R}{v} = T$ (t f

$\omega \equiv$ angular velocity; we will discuss in rotational motion chapter

$\omega R = v$

$$a = \frac{v^2}{R} = \frac{(2\pi R/T)^2}{R}$$

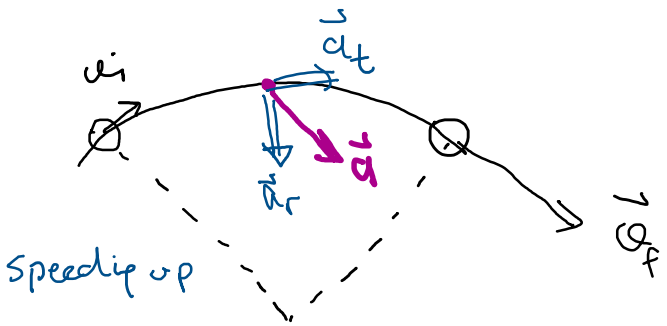
unit analys

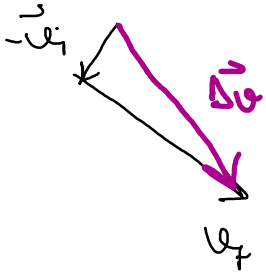
$$\frac{(m/s)^2}{m} = \left[\frac{m}{s^2}\right] \checkmark \qquad \left[\frac{m}{s^2}\right] =$$

$$a_r = -\frac{v^2}{R}\hat{r} \qquad \text{UCM}$$

If $v = \text{const} = |\vec{v}|$ $\qquad$ NUCM

If you are speeding up/ slowing down on a circular path



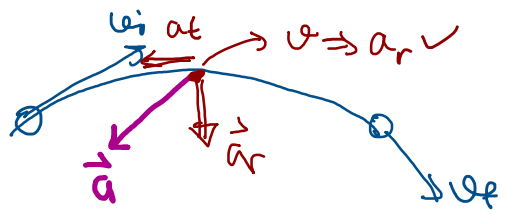

$$\frac{\Delta \vec{v}}{\Delta t} = \vec{a}$$

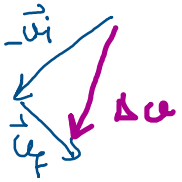
$a_t$ = tangentia accele

= (tegetse

$$\frac{\Delta|\vec{v}|}{\Delta t} = a_t = \frac{d|\vec{v}|}{dt} \quad \{ \text{derive sp}$$

If $\vec{v}(t) = At^2 + Bt \Rightarrow \frac{dv}{dt} = 2At + B \Rightarrow a_t$

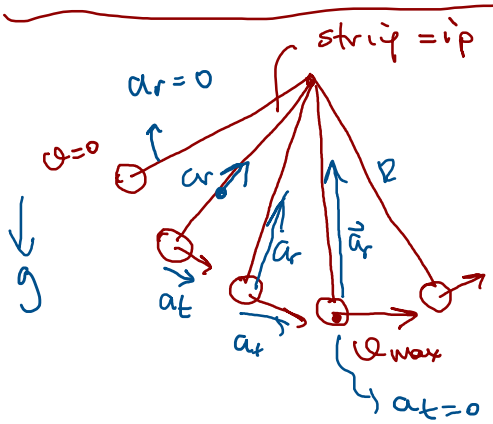



slow down

NUCM

$a_r = \frac{v^2}{R}$



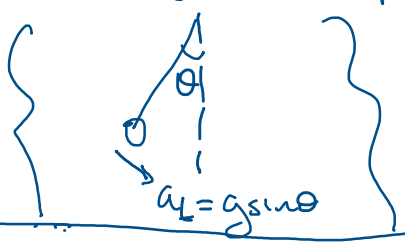

Pendulum (sarkaç) → NUCM

$a_r = \frac{v^2}{R}$

$|\vec{v}| \neq$ const

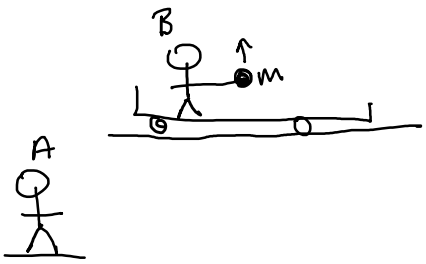
$a_t = \frac{d|\vec{v}|}{dt}$

?! $a_t = g\sin\theta$ (we'll prove it later)



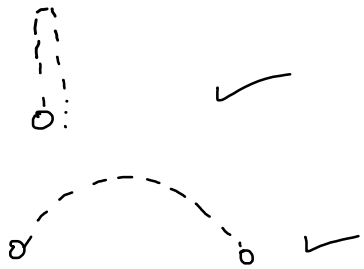

## RELATIVE MOTION (BALL MARKET)

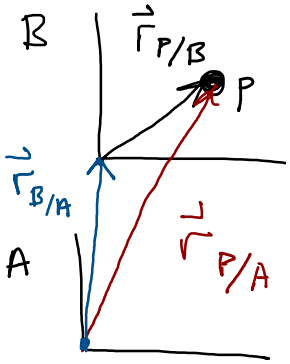
↳ 1d, 2d, 3d




acc. B

acc A





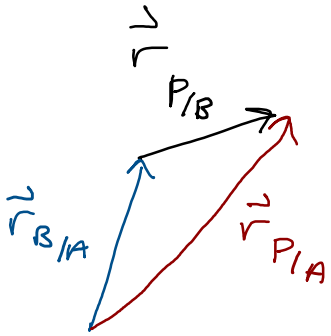

show direction of vector

$P/A$ = point P with respect to A

$P/B$ = " w. r. t. B

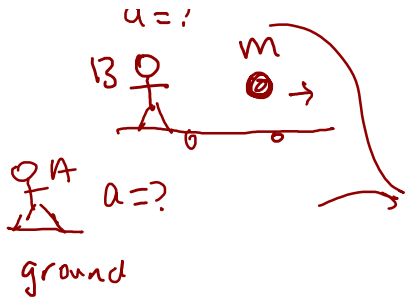
$B/A$ = cor. B w. r. t A

$A/B$ = " A w. r. t. B




$$\vec{r}_{P/B} + \vec{r}_{B/A} = \vec{r}_{P/A} \xrightarrow{d/dt}$$

$$\boxed{\vec{v}_{P/B} + \vec{v}_{B/A} = \vec{v}_{P/A}}$$




$\frac{d}{dt}$

$$\vec{a}_{P/B} + \vec{a}_{B/A} = \vec{a}_{P/A}$$

$$\vec{a}_{P/B} = \vec{a}_{P/A} \Rightarrow \underbrace{a_{B/A} = 0}_{\text{non inertial frame}}$$

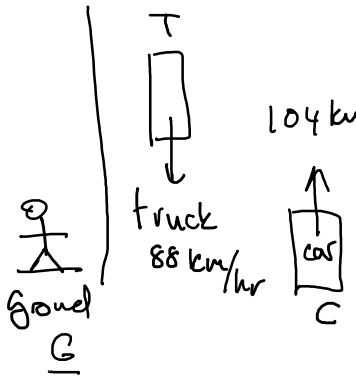
all measurements should be done on non inertial frames $\Rightarrow$ If you are measu

you cannot have

$$\vec{v}_{B/A} = -\vec{v}_{A/B}$$

head

tail




$$\vec{v}_{T/G} + \vec{v}_{G/C}$$

$\vec{v}_{C/G}$ ; $v_{C/T}$ $v_T$

$$\vec{v}_{C/T} = \vec{v}_{C/G}$$

what's the vel. of Truck w.r.t. car?

$$\vec{v}_{T/C} = ?$$

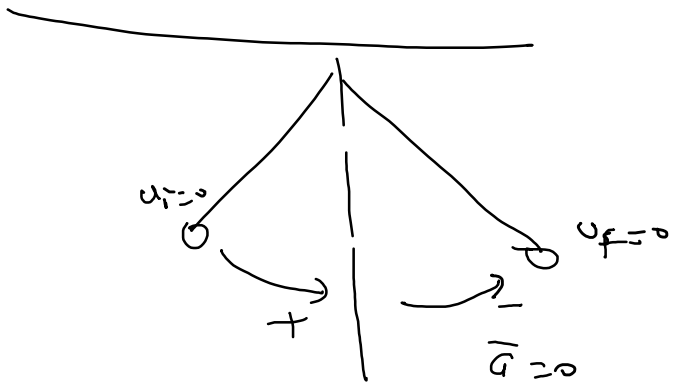
$\hat{j}$

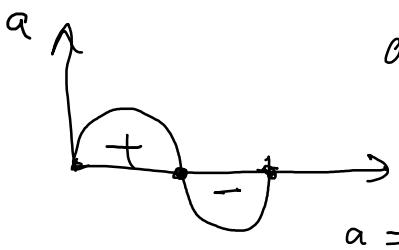
$$\vec{v}_{T/C} = \vec{v}_{T/G} + \vec{v}_{G/C}$$

$$\vec{v}_{C/G} = \uparrow +104 \text{ km/hr}$$

$$= 88\,(-\hat{j}) + (-\vec{v}_{C/G})$$

$$-88\hat{j} \quad -104\hat{j} = -192 \text{ km/hr}\,\hat{j}$$




$a_t = g\sin\theta$

$a$

$a_t$

$$a = 0 = \frac{\Delta v}{\Delta t} = v_f - v$$

$a_t = \alpha$

$v = \omega$

**Phys 101 Midterm 1** H. Beker **Fall '97**

***Question* 3 :** The velocity of a point moving along the x- axis is given by : $v(x) = \frac{A}{x}$. (everything is in MKS

a) Unit of the constant A =? , b) a(x) = ? , c) x(t) = ? If x = 0 at t = 0 , d) v (t) = ? , e) a (t) = ?

a) $v(x) = \frac{A}{x}$ unit analysis $\rightarrow \left[\frac{m}{s}\right] = \frac{A}{[m]} \Rightarrow A = \left[\frac{m^2}{s}\right]$

b) $\frac{d}{dt} v = a \Rightarrow \frac{d}{dt}\left(\frac{A}{x}\right) = A \frac{d}{dt}(x^{-1}) = A(-1)x^{-2}\left(\frac{dx}{dt}\right) = \frac{-A}{x^2} v = -\frac{A}{x^2}\frac{A}{x} = -\frac{A^2}{x^3} = a$

c) $x(t=0) = 0$ $\quad v(x) = \frac{A}{x} \Rightarrow \frac{dx}{dt} = \frac{A}{x} \Rightarrow x\,dx = A\,dt \Rightarrow \int_{x_i}^{x_f} x\,dx = \int_{t_i}^{t_f} A\,dt$

$\Rightarrow \frac{x^2}{2}\Big|_{x_i = x(0)}^{x_f = x(t)} = A\,t\Big|_{t=0}^{t=t} \Rightarrow \frac{x^2(t)}{2} - \frac{x^2(0)}{2} = At \Rightarrow x^2(t) = 2At$

$x(0) = 0$

$x(t) = \sqrt{2At}$

d) $v(t) = ?$ $\quad v(x) = \frac{A}{x} = \frac{A}{\sqrt{2At}} = v(t) = \sqrt{\frac{A}{2}}\, t^{-1/2} = v(t)$

same!

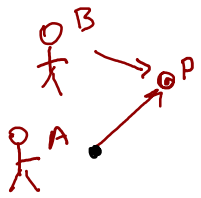
or $x(t) = \sqrt{2A}\, t^{1/2} \Rightarrow v(t) = \frac{d}{dt}x = \sqrt{2A}\,\frac{1}{2}t^{-1/2} = \sqrt{\frac{A}{2}}\, t^{-1/2}$

e) $a(t) = ?$ $\quad -\frac{A^2}{x^3} = a(x) \Rightarrow x(t) = (2A)^{1/2} t^{1/2} \Rightarrow a(t) = -\frac{A^2}{(2A)^{3/2} t^{3/2}} = \frac{-\sqrt{A}}{2\sqrt{2}\, t^{3/2}}$

or $v(t) = \sqrt{\frac{A}{2}}\, t^{-1/2} \Rightarrow \frac{dv}{dt} = a = \sqrt{\frac{A}{2}}\, -\frac{1}{2} t^{-3/2} = -\frac{\sqrt{A}}{2\sqrt{2}} t^{-3/2}$ same

## Relative motion

3 key actors $\quad \vec{v}_{P/A} = \vec{v}_{P/B} + \vec{v}_{B/A}$




$\vec{v}_{P/A}$ — head, A → tail

$\rightarrow v \quad v_{C/G}$

Ground = Earth

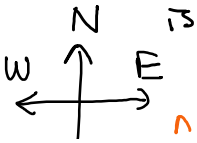
$\rightarrow v_{B/W}$

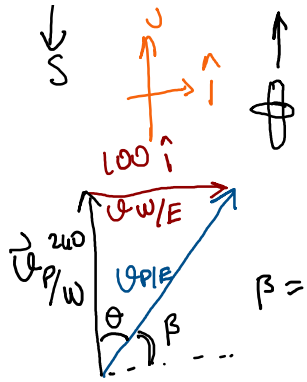
water = drift

✈ plane $\quad \vec{v}_{P/W}$

air = wind

ex) a plane's compass (pusula) shows North direction its speed is 240 km/hr. wind is west to East; wind speed is 100 km/hr. what's the velocity of plane w.r.t. Earth?






$\uparrow v_{P/W} = 240\ m/s$ $\longrightarrow v_{W/E} = 100\ km/hr$

$$\vec{v}_{P/E} = ? = \vec{v}_{P/W} + \vec{v}_{W/E}$$

$$= 240\,\hat{j} + 100\,\hat{i} \checkmark$$

$$\beta = \tan^{-1}\frac{240}{100} = 67^\circ$$

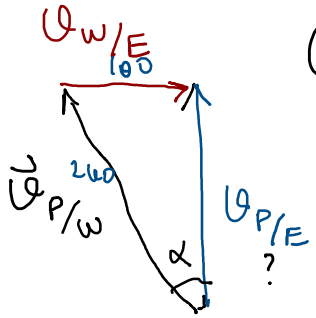
$$\left|\vec{v}_{P/E}\right| = \sqrt{240^2 + 100^2} = 260\ km/hr$$

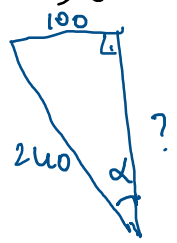
$$\theta = \tan^{-1}\left(\frac{100}{240}\right) \cong 22.6^\circ \simeq 23^\circ$$

b) Keeping the situation the same but plane aims to travel North. What would be the direction of the plane?

$$\underbrace{\vec{v}_{P/E}}_{?} = \underbrace{\vec{v}_{P/W}}_{\text{magnitude } 240} + \vec{v}_{W/E} \longrightarrow 100\,\hat{i}$$

$$(\ldots\ldots)\hat{j} = 240 \checkmark$$




$$\sin\alpha = \frac{100}{240}$$

$$\alpha = \sin^{-1}\left(\frac{100}{240}\right) = 24.6^\circ$$

$$\left|\vec{v}_{P/E}\right| = ? \sqrt{240^2 - 100^2} = 218\ km/hr$$

or $v_{P/E} = 240\cos\alpha = 218\ km/hr$

End of Ch 3